



Vaishnavi Shirbhate

Machine Learning / Computer Vision

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I am M.Sc. Computer Science graduate with a focus on Artificial Intelligence and experience in computer vision, machine learning, and applied research across medical imaging and automotive perception. Experienced in developing and evaluating deep learning methods for medical image reconstruction, working with large-scale imaging data, and building reproducible ML pipelines using Python and PyTorch. Particularly interested in applying machine learning to medical imaging and developing image reconstruction methods that can support faster and more practical clinical workflows. English (C2), German (A2)

TECHNICAL SKILLS

- **Programming:** Python, SQL
- **Computer Vision & ML:** Deep Learning, Computer Vision, Generative Modelling, Transformers, PEFT, Object Detection, Image Processing, CNNs, GANs, Diffusion Models, Model Benchmarking, Quantitative Evaluation
- **Libraries & Frameworks:** PyTorch, PyTorch Lightning, Pandas, NumPy, OpenCV, Matplotlib, MLFlow
- **Developer Tools:** Git, GitHub, Weights & Biases, Linux, Databricks, Slurm/HPC, Jira

RESEARCH & PROJECTS

02/2026 – Present

Master Thesis | AI-Driven Image Restoration for RSOM Skin Disease Diagnostics | DKFZ

- Improved reconstruction of Photoacoustic mesoscopy data by developing an end-to-end deep learning pipeline in PyTorch Lightning for medical image restoration.
- Benchmarked three classes of reconstruction methods by implementing and comparing CNN and Transformer based regression models, GAN, and diffusion-based architectures across the same imaging task.
- Established quantitative assessment of reconstruction quality by evaluating image and medical domain metrics.
- Improved understanding of reconstruction limitations by analysing model failure cases across different image structures and vessel sizes.
- Compared two reconstruction formulations by evaluating image-domain and data-domain restoration approaches to determine their effect on reconstruction quality.
- Improved reproducibility of reconstruction experiments by implementing structured experiment tracking, hyperparameter management, and model comparison using Weights & Biases.

04/2025 – 10/2025

Master Project | Sign Language Image Generation (Generative AI) | RPTU

- Benchmarked multiple deep generative vision approaches by implementing SwinGAN, and Latent Diffusion Models in PyTorch and comparing their image-generation performance.
- Reduced inference latency from 163 ms to 16 ms (10×) by developing a custom latent diffusion pipeline with an optimised inference process.
- Enabled training on a 3 TB image dataset by developing scalable preprocessing and data-loading pipelines for large-scale image data.
- Improved evaluation of generative image models by combining quantitative metrics with visual analysis across different architectures.

04/2024 – 10/2024

LLM Fine-tuning for Code Generation | RPTU

- Improved code-generation model quality to 85% by fine-tuning LLaMA with parameter-efficient fine-tuning and curated synthetic Python data.
- Expanded the training data available for fine-tuning by generating and processing synthetic Python code covering additional programming patterns.

10/2024 – 02/2025

Seminar | Camera-LiDAR Fusion for 3D Object Detection in Autonomous Driving | RPTU

- Analyzed camera-LiDAR fusion approaches for 3D perception by studying sensor calibration, multimodal representations, BEV methods, and transformer-based architectures.
- Compared state-of-the-art 3D perception methods using nuScenes benchmark metrics to evaluate detection accuracy and localization performance.

PROFESSIONAL EXPERIENCE

09/2024 – 02/2025 Kaiserslautern	Research Assistant German Research Center for Artificial Intelligence (DFKI) - Improved the reliability of a 40+ hour medical video dataset by developing annotation guidelines, validating labels, and introducing structured quality-control procedures for computer vision research. - Reduced manual preparation of multimodal medical data by developing an end-to-end synchronisation pipeline for aligning video and sensor streams into a unified dataset. - Enabled repeatable processing of large-scale medical data by automating preprocessing and dataset generation workflows for computer vision experiments. - Improved data quality for downstream model development by systematically identifying and resolving inconsistencies in annotations and synchronised data.
06/2024 – 02/2025 Kaiserslautern	Student Assistant & Course Tutor, Machine Learning Group RPTU - Developed diffusion-based models for spatio-temporal data generation by implementing and evaluating different diffusion architectures in PyTorch. - Improved the repeatability of model evaluation by running controlled experiments across different model configurations and training settings. - Supported the learning of 30+ students by teaching concepts including transformers, reinforcement learning, NLP, and LLMs and evaluating technical assignments.
06/2018 – 08/2021 India	Systems Engineer, Computer Vision Tata Consultancy Services - Improved real-time object detection for automotive camera systems by developing and optimising YOLO-based computer vision pipelines for ZF Friedrichshafen. - Reduced inference latency under computational constraints by optimising preprocessing and deployment components while maintaining detection performance. - Improved reliability of computer vision components by debugging and testing perception pipelines throughout development. - Delivered 20+ production software releases by developing and integrating C++ components within Agile and CI/CD workflows for automotive systems.

EDUCATION

10/2023 – Present Germany	M.Sc. Computer Science, RPTU Kaiserslautern-Landau (Grade: 2.0) - Focus: Intelligent Systems (AI) · Visualisation & Scientific Computing - Member of Machine learning group and Theater Club
08/2014 – 05/2018 India	B.E. Computer Science and Engineering, Amravati University (PRMITR) (Grade: 1.7)